

Rešitve

3. Izpit
2. kolokvij

24.8.2020 (odpadel)
26.1.2021

2. naloga



Postavimo $T_0 = 0$ (Oziroma $T_{\text{emp}} = T(\theta, t) + T_0$)

Začetni . p. $T(\theta, 0) = C_0 \delta(\theta - \frac{\pi}{2})$; C_0 povezan z Q kot: $Q = \int_{\text{sv. d.}} 2\pi r \cdot r d\theta T_{\theta=0} = C_0 c_p d 2\pi r^2 \theta_0$?

Robni . p. $\frac{\partial T}{\partial \theta} \Big|_{\theta=\frac{\pi}{2}} = 0$

$$C_0 = \frac{Q}{c_p d 2\pi r^2 \theta_0}$$

1t

Rešujemo Laplace $\nabla^2 T = \frac{\partial T}{\partial t}$

$$\Rightarrow T(\theta, t) = \sum_{l=0}^{\infty} A_l P_l(\cos \theta) e^{-\frac{l(l+1)}{r^2} Dt}$$

1t

$$\Rightarrow \frac{\partial T}{\partial \theta} \Big|_{\theta=\frac{\pi}{2}} = 0 \Rightarrow \frac{\partial P_l}{\partial \theta} \Big|_{\cos \theta = 0} = 0 \Rightarrow \text{sodi } l$$

1t

$$T(\theta, 0) = C_0 \delta(\theta - \frac{\pi}{2}) = \sum_{l=0}^{\infty} A_{2l} P_{2l}(\cos \theta)$$

$$C_0 P_{2g}(0) = A_{2g} \frac{2}{4g+1} \cdot \frac{1}{2}$$

Važna polovička!
(Preverimo za $l \rightarrow \infty$)

Poi
stene!

$$\int_{-1}^1 dx P_g(x) dx =$$

$$\int P_g P_l = 2\delta_{gl} \cdot \frac{1}{2}$$

$$\Rightarrow A_{2g} = \frac{4g+1}{2} C_0 \cdot P_{2g}(0)$$

$$A_0 = C_0 \cdot 1$$

$$P_{2g}(0) = \frac{(-1)^g}{4^g} \binom{2g}{g}$$

ni jim
treba
vedeti
tega

$$T(\theta, t) = \sum_{g=0}^{\infty} \frac{4g+1}{2} C_0 P_{2g}(0) P_{2g}(\cos \theta) e^{-\frac{2g(2g+1)}{r^2} Dt}$$

2t

$$t \rightarrow \infty$$

$$T = C_0 = \frac{Q}{c_p (d 2\pi r^2) \theta_0}$$

Pravilno

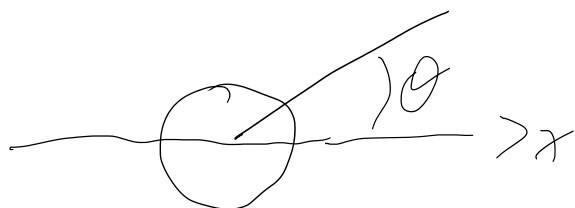
Če ni $\frac{1}{2} \rightarrow -0.5t$

Positive 2. kol 4. naloga

Tuesday, 26 January 2021

21:47

a)



$$\Phi = \Phi_i + \Phi_s \quad \Phi_i = e^{ikx - i\omega t}$$

$$\Phi_s = \sum_{n=0}^{\infty} A_n H_n^{(1)}(kr) \cos n\theta e^{-i\omega t}$$

1

$$RP: \left. \frac{\partial}{\partial r} (\Phi_i + \Phi_s) \right|_{r=R} = 0$$

$$= \sum_{n=0}^{\infty} A_n H_n^{(1)'}(kR) k \cos n\theta + k J_0'(kR) + \sum_{n=1}^{\infty} 2i^n J_n'(kR) \cos n\theta k$$

1

$$A_0 = -\frac{J_0'(kR)}{H_0^{(1)'}(kR)}$$

1

$$A_n = -2i^n \frac{J_n'(kR)}{H_n^{(1)'}(kR)}$$

$$b) \text{ Tlak } S_p = \rho \frac{\partial \Phi}{\partial t} = -i\omega \rho e^{-i\omega t} (e^{ikx} + \sum A_n H_n \cos n\theta)$$

Je simetrije vidimo, da bo horizontalna b x komponenta sila (projekcija na x-komp)

1

$$\frac{F}{dL} = \int_0^{2\pi} S_p(\theta) \cdot (-\cos \theta) R d\theta$$

$$\text{uporabimo } \int_0^{2\pi} \cos n\theta \cos \theta d\theta = \delta_{n,1} \pi$$

$$\frac{F}{dL} = +R i \omega \rho e^{-i\omega t} \pi \left(2i J_1(kR) + A_1 H_1^{(1)'}(kR) \right)$$

1

$$= R \omega \rho \pi e^{-i\omega t} \left(-2 J_1(kR) + 2 \frac{J_1'(kR)}{H_1^{(1)'}(kR)} H_1^{(1)'}(kR) \right)$$